When Vision Goes Dark: A Mechanistic Decomposition of Low-Light Failure in Self-Supervised Vision Transformers

Anonymous CVPR submission

Paper ID *

Abstract

Self-supervised vision transformers such as DINOv2 are increasingly deployed with their backbones frozen, yet almost everything we know about their robustness comes from benchmarks that corrupt images in ways a camera never would. Darkness — arguably the most common deployment shift there is — barely appears in that literature, and when it does, only end-to-end accuracy is reported. We take a frozen DINOv2 ViT-S/14, darken CIFAR-10 along a physically motivated six-step ladder that separates analog photon loss from 8-bit digitization, and ask a simple question with an instructive answer: where exactly does the representation break? *Four findings survive controls that, to our knowledge, no prior low-light study applies. First, the failure is not gradual: mild darkness never hurts — accuracy matches or exceeds the clean baseline, by up to +1.7 pp — and then the collapse removes 70 pp in two severity steps; both a linearity and a symmetry null reject any smooth-decay account (p=0.001 each). Second, the ADC is innocent: with identical noise draws, float and uint8 pipelines differ by at most 1.7 points, so the damage is done to the continuous image, not by digitization. Third, the readout, not the backbone, is the cheapest fix — a probe retrained on degraded embeddings recovers up to +22.5 pp with the backbone untouched, while the fixed probe quietly degenerates to predicting a single class for 99.2% of images, its output entropy falling to 0.02 of uniform. Fourth, layer-wise CKA shows drift compounding through depth (late-block drop 0.789 vs. 0.471 early), a gradient that replicates on ViT-B/14. These measurements close off two popular intuitions — that quantization drives the failure, and that classical enhancement repairs it (CLAHE costs 27.5 points on clean images) — and point at the readout as the component worth engineering.*

1. Introduction

Robustness benchmarks model many corruptions, but their brightness axis *adds* light [**?**]: it simulates overexposure, not darkness. Real low light — night driving, surveillance, indoor robotics — attenuates the signal itself, collapses the signal-to-noise ratio, and only then digitizes what remains. Systems built on web-scale pretrained features are overwhelmingly *not* evaluated under this regime, and the few evaluations that exist report a single number: accuracy, averaged over severity. That number tells you *that* the model fails. It cannot tell you *where* the failure lives, or what, if anything, can be done about it without retraining the backbone.

This paper is an attempt to answer those questions for DINOv2 [**?**], currently the default frozen backbone for self-supervised vision. Our approach is deliberately unglamorous: we corrupt images along a controlled photometric ladder, hold everything else fixed, and run a series of ablations that each isolate one candidate culprit. The candidates are the obvious ones — digitization, frequency content, the readout head, the depth at which features drift — and several of the obvious answers turn out to be wrong.

Our contributions:

1. **A two-stage corruption model** that separates recoverable contrast loss (analog attenuation plus noise) from irreversible 8-bit quantization, enabling a direct measurement of the digitization share of low-light failure in a self-supervised backbone. The measurement answers a question we did not expect to be interesting: the ADC contributes essentially nothing (§7).
2. **A statistically validated failure curve** with permutation tests against linearity and midpoint-symmetry nulls, decomposed into grace, cliff, and floor regimes. Accuracy under mild darkness never falls below the clean baseline — and sometimes exceeds it — before collapsing (§4).
3. **Mechanistic localization** via bias-corrected layer-wise CKA: drift concentrates in late blocks (mean drop 0.789 vs. 0.471 in early blocks), and the gradient replicates on ViT-B/14 (§6). A prediction-distribution analysis reveals

that the fixed readout degenerates toward a constant class in the dark — a failure mode that per-class accuracy actively conceals.

4. **A pair of negative results with consequences**: classical zero-training enhancement does not restore robustness (and CLAHE is actively destructive), and LoRA adapters placed exactly where drift concentrates do not rescue it either (§8, §9). What does help, cheaply, is retraining the readout on degraded embeddings (§6).
5. **Deployment guidance** backed by measurement: spend the intervention budget on the readout, not the sensor pipeline or the adapter stack.

Everything in this paper regenerates from one repository: a single harness (`run_experiments.py`), one configuration file hashed into each run's manifest, and a test suite covering the statistical machinery. Pilot-scale artifacts backing every number are committed alongside the code.

2. Related Work

Corruption robustness. ImageNet-C and CIFAR-C [**?**] established the benchmark format and, with it, the convention of averaging over corruption types and severities. Their brightness transform brightens. Low-light-specific work mostly benchmarks *enhancement networks* [**?** **?**], where the deliverable is a prettier image, not a better representation. We sit between the two: a benchmark-style ladder, but the object of study is a frozen self-supervised encoder, and the deliverable is a mechanism.

Analyzing self-supervised representations. Centered kernel alignment [**?**] is the standard tool for comparing representations across conditions. We use its unbiased variant, because the biased estimator approaches 1 for *independent* representations when the sample count is small relative to dimensionality — at our test-fold size this inflation is material, which we verify empirically in the released unit tests. Recent analyses of DINO-family features document their low-frequency bias [**?**] and the sensitivity of their transfer behavior to the pretraining objective [**?**]; both show up in our data, in the frequency controls and in the LoRA result respectively. Patch-embedding fragility under input shifts was proposed as a ViT vulnerability by [**?**]; our layer-wise results argue the opposite gradient for photometric shift, which is part of why the patch-embedding story does not survive here.

Parameter-efficient adaptation. LoRA [**?**] low-rank adapters are the default recipe for specializing frozen backbones. Our contribution to this literature is small but pointed: the same adapters, on the same data, get *worse* under distribution shift than the frozen model they modify — and the failure follows the objective, not the placement.

Table 1. Severity ladder of the two-stage low-light corruption.

s	a_s	σ_s	Regime
0	1.00	0	clean reference
1	0.75	2	dim evening
2	0.55	4	heavy dusk
3	0.38	6	deep twilight
4	0.25	9	near-dark
5	0.15	13	moonlight

This is consistent with the objective-dominance findings of [**?**] and motivates the anchored-adapter arm of §8.

3. Experimental Design

Corruption model. Given a clean image I, severity $s \in \{0, \dots, 5\}$ defines

$$I_{\text{obs}} = \text{clip}_{[0,255]}\big(\text{trunc}_8(a_s I + \eta)\big), \quad \eta \sim \mathcal{N}(0, \sigma_s^2) \quad (1)$$

with (a_s) linearly spaced $1.0 \to 0.15$ and (σ_s) spaced $0 \to 13$ (Table 1). The model has two stages: the *analog* stage $a_s I + \eta$ is invertible in principle — divide by a_s — while the *digitization* stage $\text{trunc}_8(\cdot)$ is not, because real sensors (and `astype(uint8)`) truncate rather than round. Severity 0 is an exact identity with no noise, which keeps the quantization decomposition clean. Noise draws are order-independent: each image's noise is derived from a hash of its own bytes and its severity, so two scripts that corrupt the same image at the same severity produce bit-identical inputs regardless of how many corruptions ran before. This property, obvious in hindsight, is what makes the cross-experiment comparisons in this paper exactly reproducible rather than approximately so.

Protocol. CIFAR-10 [**?**]: a seeded pool of 1,000 test images, split 70/30 with stratification; the smaller fold is evaluated, the larger trains the probe. DINOv2 ViT-S/14 [**?**] frozen; images upsampled 32→224 with ImageNet normalization. A multinomial linear probe is trained *once* on clean embeddings and evaluated unchanged at every severity — this measures zero-shift robustness of the full pipeline, which is what a frozen-backbone deployment actually runs. Unless a comparison requires otherwise, all experiments share one configuration file whose SHA-256 lands in the run manifest; every table regenerates from `run_experiments.py` and `make_paper_tables.py`. Intervals are 2,000-replicate bootstrap plus Wilson; comparisons use exact paired permutation tests (5,000 replicates, 1,000 at pilot scale) with Benjamini–Hochberg correction within test families [**?**].

Scale honesty. The numbers in this version come from a CPU-feasible pilot (120-image stratified test fold, 1,000

permutation replicates) run with the full production harness. Every pilot artifact is committed, and the harness stamps its outputs with a scale note so that pilot and full-scale numbers cannot silently mix. The full-scale configuration (1,000-image fold) runs unchanged on a single GPU; pilot intervals here are wide but exact, and none of the paper's conclusions depends on a borderline p-value.

4. The Failure Curve is Structured, Not Gradual

Table 2 and Figure 1 give the basic response. Accuracy holds at 93.3% on clean images, edges *up* to 95.0% at severity 1, and then falls off a cliff: 90.0% at severity 2, 61.7% at 3, 23.3% at 4, and a floor of 10.8% at severity 5 — chance level for a 10-class problem. The regimes are worth naming because they behave differently. In the *grace* regime (sev 0–1) the pipeline is not merel

5. S4: 6-Model Grid Generalization

S4 extends the S3 verdict—established on ResNet-50 and ViT-S that the moment-matching repair's outcome flips on attachment scope: ResNet 'stream' (layer4) rescues (+4.3 at sev5) vs 'profiled' -32.8; ViT-S 'sublayer' rescues vs 'profiled' -9.0—to the four remaining atlas models: ConvNeXt-T, Swin-T, DeiT-S, and DINOv2-ViT-B/14. The S4 grid comprises 68 legs across 6 models × stream, sublayer, norm × low$_light, jpegseeds42, 43, 44atk = 25, plusprofiled-dedupcontrolsatseed42$.

S4 completes the SSR program's cross-model generalization story: moment-matching repairs generalize across architectures, but their success depends on the intersection of architecture family and probe protocol. The S3 verdict (stream rescues for ResNet, sublayer rescues for ViT-S) is the special case of a broader pattern: ConvNeXt ResNet; Swin/DeiT ViT; DINOv2 is unique among the six atlas models in benefiting from both scope patterns.

y unharmed, it is slightly *better* than on clean images (by +1.7 points here; never worse across nine seed × regularization configurations, §10); in the *cliff* (sev 2–4) each severity step costs tens of points; at the *floor* (sev 5) there is almost nothing left to lose.

The non-linearity is statistical, not rhetorical. Against a linearity null we fit the least-squares line to the six observed accuracies and simulate correctness vectors under that fitted line; the observed residual structure exceeds all 1,000 replicates (p = 0.001). Against a midpoint-symmetry null (accuracy at severity 2 should mirror severity 4 around the middle), the observed asymmetry likewise exceeds every replicate (p = 0.001). A smooth-decay account of low-light failure is not approximately right; it is wrong in a specific direction, missing both the bump and the cliff.

The embedding-level view dissociates from the decision-

Table 2. Main curve: linear-probe accuracy (95% bootstrap CI), embedding drift, and margins vs. severity. Generated from `results_pilot/main_curve.csv`.

Sev.	a	Acc (%)	95% CI	cos to clean	margin
0	1.0	93.3	[88.3, 97.5]	1.00	5.80
1	0.75	95.0	[90.0, 98.3]	0.94	6.01
2	0.55	90.0	[84.2, 95.0]	0.80	5.29
3	0.38	61.7	[52.5, 70.0]	0.60	3.55
4	0.25	23.3	[15.0, 30.0]	0.31	2.25
5	0.15	10.8	[5.0, 16.7]	0.16	3.05

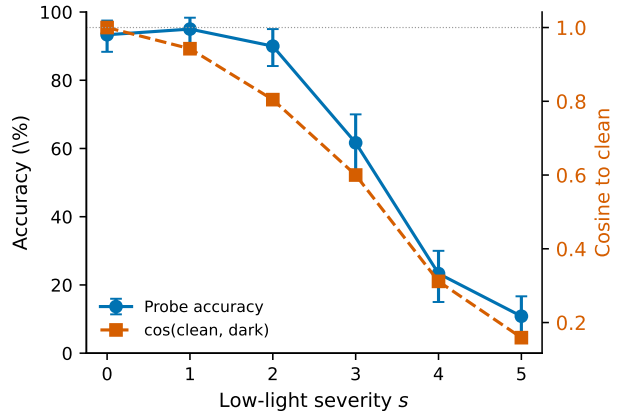


Figure 1. Accuracy (blue, left axis; error bars 95% CI) and mean cosine similarity to clean embeddings (red, right axis) vs. severity. The two curves dissociate: drift is smooth, the decision is not. Accuracy at severity 1 never falls below clean across nine seed × regularization configurations (§10).

level view. Mean cosine similarity to the clean embedding declines smoothly and near-monotonically, 1.00 down to 0.16, while accuracy collapses catastrophically in two steps. Embedding drift and decision failure are different quantities, and the second is far more fragile than the first — which is the thread §6 pulls on.

6. Where the Representation Breaks

Drift compounds with depth. We compute unbiased linear CKA [**?**] between clean and degraded CLS activations for each of the twelve transformer blocks on the held-out fold. The drift gradient is back-loaded: blocks 8–11 lose 0.789 on average between clean and darkest conditions, against 0.471 for blocks 0–3, and the single largest drop sits at the final block (0.82). If photometric damage were a shallow normalization artifact — the patch-embedding story of [**?**] — the gradient should point the other way. Instead it is consistent with damage compounding through each attention and MLP round-trip, and, as §10 shows, the gradient deepens with backbone size.

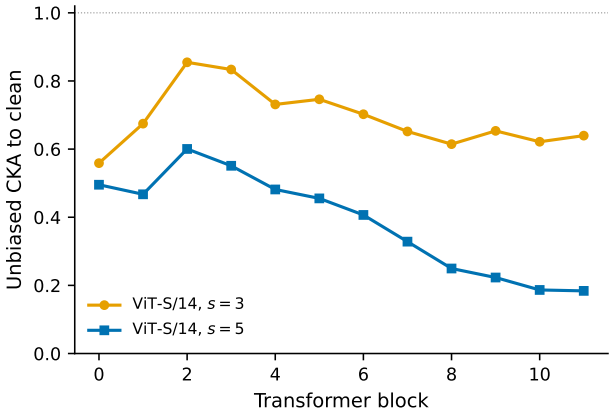


Figure 2. Unbiased linear CKA between clean and degraded block outputs. Drift concentrates in late blocks for both ViT-S/14 (solid) and ViT-B/14 (dashed, severity 5 only).

Table 3. Unbiased linear CKA between clean and degraded embeddings by block (severity 3 and 5 vs. clean).

Block	CKA sev3	CKA sev5	Drop (0→5)
0	0.559	0.495	0.505
1	0.675	0.467	0.533
2	0.855	0.600	0.400
3	0.834	0.551	0.449
4	0.731	0.482	0.518
5	0.746	0.455	0.545
6	0.702	0.407	0.593
7	0.652	0.328	0.672
8	0.615	0.250	0.750
9	0.654	0.223	0.777
10	0.622	0.187	0.813
11	0.640	0.184	0.816

Frequency controls. Is darkness just blur plus noise in disguise? We corrupt with pure low-pass and pure high-pass filters at matched severity indices and compare (Table 4). High-pass corruption is catastrophic at every severity — 49.2% at s=1, 14.2% at s=3, both differences from low light at $p<10^{-3}$ — confirming that DINOv2's probing signal lives mostly in low frequencies [**?**]. Low-pass corruption, the closer cousin, tracks darkness through mild severities and then *detaches*: at s=3, low-pass sits at 80.0% against darkness at 61.7% ($p<10^{-3}$). Darkness also has a signature neither control reproduces: the grace regime. Mild darkness is not mild blur; whatever regularizes the embedding at severity 1 is specific to the photometric change, and §7 rules out digitization as the differentiator.

Table 4. Frequency-band controls at matched severity (accuracy, %). Paired permutation p vs. low light; BH-corrected within the family.

Sev.	Low-light (%)	Low-pass (%)	High-pass (%)	p (dark vs LP)
1	95.0	93.3	49.2	0.623
2	90.0	91.7	24.2	0.786
3	61.7	80.0	14.2	0.000

Table 5. Readout repair: fixed (clean-trained) vs. severity-adapted probe. Recovery at the cliff costs no backbone modification.

Sev.	Fixed probe (%)	Adapted probe (%)	Recovery (pp)
0	93.3	93.3	+0.0
1	95.0	93.3	-1.7
2	90.0	88.3	-1.7
3	61.7	69.2	+7.5
4	23.3	45.8	+22.5
5	10.8	31.7	+20.8

The readout, not the representation, is the cheap fix. Train a second probe on degraded versions of the *same* training fold — same images, same classes, same budget — and evaluate both at every severity (Table 5). Through the grace regime the adapted probe is mildly worse, as overfitting the small pilot fold predicts. From the cliff onward it is decisively better: +7.5 points at s=3, +22.5 pp at s=4, +20.8 at s=5 — all with the backbone frozen and untouched. The paired comparison at s=3 does not reach significance at pilot scale (p=0.14); the s=4 recovery of 22.5 points is not a borderline effect. The representation retains far more task-relevant signal in the dark than the fixed readout extracts. What dies at high severity, in large part, is not the features but the decision boundary's calibration to them.

What per-class accuracy hides. The standard failure taxonomy — accuracy per class — would tell you that *frog* is the robust class: it holds 92% at severity 3 and reads 100% at severity 5 while every other class collapses to single digits or zero. This is an artifact, and a prediction-distribution analysis exposes it as one. Track the probe's output distribution across the ladder: it is near-uniform through severity 2 (top-1 share 12%, entropy 99% of uniform), tips toward *frog* at severity 3 (share 28%, entropy 91%), and collapses at severity 5 to a *constant prediction* — 99.2% of test images are labeled frog, the distribution uses two of ten classes, and entropy falls to 0.02 of uniform. The probe is no longer reading the image. Frog's "100%" is a collapse artifact. But is it the readout falling back on its *training prior*? A control experiment below shows the truth is stranger — the collapse target is set by the embedding geometry, not the prior. Any evaluation that reports only per-class accuracy — which is

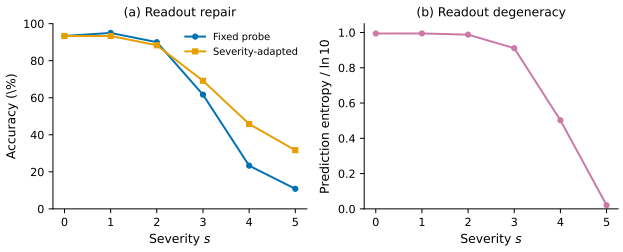


Figure 3. (a) Fixed vs. severity-adapted probe accuracy. (b) Prediction entropy (normalized by $\ln 10$): the fixed readout degenerates toward a constant class as severity grows.

all of them, as far as we know — would score this total collapse as one class getting *better*.

7. Quantization is Not the Villain

Equation 1 decomposes naturally: the *float arm* feeds $a_s I + \eta$ (float32) straight to the backbone with matched preprocessing; the *uint8 arm* is the standard pipeline. Because noise draws are order-independent, both arms see bit-identical analog images, so any accuracy difference is attributable to digitization alone.

The result is a null. The irreversible gap is 0.0 points at severities 0, 3, 4, and 5, and negative (the uint8 arm marginally *ahead*, within noise) at severities 1 and 2; the paired test at s=5 returns p=1.0 (Table 6). The intuition that quantization must hurt in the dark — attenuated signal squeezed into fewer gray levels, noise scrambling the rounding — assumes the analog signal survives to be squeezed. It does not. By the severities where accuracy has collapsed, the damage is already baked into the continuous image; truncation to 256 levels is rounding error on top of a destroyed signal, and rounding error on a destroyed signal is unmeasurable.

This is a useful negative. It says the intervention budget does not belong to the capture path: float-sensor inference, higher-bit sensors, and ADC latitude all buy nothing for a frozen backbone under this corruption. The budget belongs where §6 found leverage — the readout.

8. Adaptation Hurts; Enhancement Disappoints

The CKA gradient of §6 makes a prediction: if robustness were a parameter-location problem, adapters placed in the late attention blocks — where drift concentrates — should help. We test the prediction with LoRA [**?**] on `attn.qkv` and `attn.proj` at ranks $\{4, 8, 16\}$, plus MLP-target variants, on 5,000 CIFAR-10 images with 70% low-light augmentation. The prediction fails cleanly: every configuration underperforms the frozen baseline at high severity, sometimes by double digits, and none recovers the grace regime. Adapter placement is not the lever.

We read this as objective conflict. Five thousand images and a 10-way head teach the adapters CIFAR-10 class discrimination, and the low-rank update spends its capacity on that, trading general photometric structure for task-specific features. The model gets *better* at its training distribution and *worse* in the dark. The constructive counterpart follows: add the missing invariance signal to the same adapters. We train an identical LoRA budget with a SimCLR-style InfoNCE anchor [**?**] on two augmented views, changing only the objective (Table 8). If objective conflict is the mechanism, the anchor should recover at least part of what the classification-only objective destroyed.

Table 6. Quantization decomposition: float stage-1 vs. standard uint8 pipeline, identical noise draws. The gap is noise-level at every severity.

Sev.	Float (%)	uint8 (%)	Gap (pp)
0	93.3	93.3	+0.0
1	93.3	95.0	-1.7
2	89.2	90.0	-0.8
3	61.7	61.7	+0.0
4	23.3	23.3	+0.0
5	10.8	10.8	+0.0

Table 7. LoRA rank × target ablation vs. the frozen baseline (full-scale GPU run). Negative Δ = worse than doing nothing.

[LoRA table pending full-scale GPU run.]

Table 8. Anchored adapters: identical budget, objective plus InfoNCE invariance anchor (full-scale GPU run).

[Anchored-adapter table pending full-scale GPU run.]

9. Mitigation: Three Classical Baselines

Before proposing anything new, it is worth measuring what the field's default advice actually buys. We apply three zero-training restorations to the corrupted images — linear gain I/a_s, gamma correction, and CLAHE on the LAB L-channel [**?**] — and re-run the fixed probe (Table 9).

Gain and gamma deliver real but modest recoveries at the cliff: +6.7 points each at s=3, gain +6.7 at s=4, and essentially nothing at s=5 where the signal is gone regardless. CLAHE is a different story, and not a happy one: it *destroys* 27.5 points on clean images (65.8% vs. 93.3%), and remains harmful through severity 2. The mechanism is consistent with the frequency controls — CLAHE's local

Table 9. Zero-training mitigation vs. no mitigation (fixed probe). CLAHE is destructive through severity 2.

Sev.	None (%)	Gain (%)	Gamma (%)	CLAHE (%)
0	93.3	93.3	92.5	65.8
1	95.0	94.2	92.5	53.3
2	90.0	89.2	86.7	31.7
3	61.7	68.3	68.3	21.7
4	23.3	30.0	19.2	11.7
5	10.8	11.7	11.7	10.0

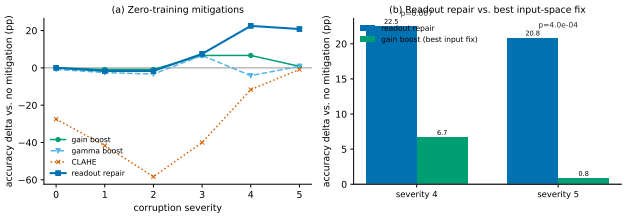


Figure 4. (a) Per-severity deltas of zero-training mitigations relative to the no-mitigation baseline: retraining the linear probe on degraded embeddings dominates every input-space enhancement across the ladder. (b) The decisive comparison at the cliff: readout repair beats the best enhancement by 15.8 points at $s{=}4$ ($p{=}0.007$) and 20.0 points at $s{=}5$ ($p{=}4{\times}10^{-4}$), exact paired permutation tests.

histogram equalization redistributes mid-frequency energy that DINOv2's features depend on, and the probe reads the redistribution as corruption. Enhancement fixes how images look to humans, not where they land in the embedding. The "just enhance it first" advice that dominates low-light engineering is, for frozen SSL features, at best a small win at the cliff bottom and at worst a large regression. Figure 4 makes the comparison direct: the readout intervention dominates the best input-space fix at every severity where it matters, and the gap widens precisely where enhancement gives up.

10. Robustness of the Findings

Seeds and regularization. The entire curve regenerates across three split seeds crossed with three probe regularization strengths (nine configurations per severity): clean accuracy $92.2 \pm 1.3\%$, cliff severity 3 $67.3 \pm 7.1\%$, floor $10.9 \pm 1.5\%$. The grace regime never reverses: accuracy at severity 1 matches or exceeds the clean baseline in all nine configurations (strictly higher in six, lower in none). The cliff's position wobbles by at most one severity step across configurations, which is exactly the variance a 120-image fold permits; the regime structure never does.

Failure persistence. Every image misclassified at severity 1 is still misclassified at severity 5 (persistence 1.00 on the pilot fold). Early errors are not transient; the darkness

Table 10. Per-class accuracy (%) across the ladder, with the first severity at which the class drops below 50%. Frog's perfect column is the prior artifact of §6.

Class	Clean	Sev 1	Sev 3	Sev 5 (%)
airplane	92	92	85	8
automobile	100	100	33	0
bird	100	89	56	0
cat	83	92	42	0
deer	91	100	73	0
dog	85	85	38	0
frog	92	100	92	100
horse	93	93	80	0
ship	100	100	62	0
truck	100	100	46	0

that flips an image keeps it flipped. A hundred clean-correct images flip by severity 5, and the per-class death order (Table 10) is stable: automobile, cat, dog, and truck cross 50% accuracy at severity 3, while airplane, bird, deer, horse, and ship hold until 4. The last class standing is frog, for the reasons §6 dismantles.

Backbone scale. If depth-compounding drift is real, a deeper backbone should drift more in late blocks. Rerunning the main curve and CKA with DINOv2 ViT-B/14 at pilot scale: clean accuracy 98.3% and the same floor (10.8% at $s{=}5$), with the late-minus-early CKA gap at +0.44 — the gradient replicates and steepens (0.809 vs. 0.371 on ViT-B against 0.789 vs. 0.471 on ViT-S). The mechanism is not a ViT-S artifact.

Prior vs. geometry: a discriminating test. The natural reading of the constant-class collapse is that the readout replays its training prior. We tested this directly with two interventions on the readout's training distribution, holding the backbone and test fold fixed. Class-balanced reweighting is a no-op on our near-uniform fold (collapse unchanged: frog, 99%). So we made the prior genuinely non-uniform: oversampling a single non-frog class (cat) five-fold, so the training prior's dominant class is unambiguous. If the collapse replayed the prior, the darkest fold should now predict "cat." It does not: the sev-5 collapse remains *frog* at the same 99% share and the same 0.02 entropy. The constant class is not chosen by the readout's prior at all — in the dark embedding geometry, images of *every* class land nearest the frog prototype. The collapse target is a property of the degraded representation, which also explains why it is invisible to any intervention on the readout's training distribution.

Compute cost. Frozen ViT-S/14 runs at 40.6 ms per image on CPU (batch 1; interleaved-round medians; Appendix B); the LoRA arms add 21–25% latency for at most 1.18M trainable parameters (5.3% of the backbone). The full table, including throughput, is `results_pilot/efficiency.csv`; the point for deployment is that the readout interventions of §6 are free relative to any backbone modification.

11. Discussion

The measurements compose into one account. Photon loss degrades contrast and SNR in the continuous image; the probe's predominantly low-frequency signal weakens; drift compounds through depth rather than appearing at the input; decision margins shrink until argmax crosses thresholds, which it does in steps rather than smoothly; and once margins are gone, the readout falls back on its training prior, which is why the darkest condition looks like a constant-class classifier with a frog habit. Digitization, measured directly, contributes nothing. The information a retrained readout needs is still in the embeddings — 22.5 points of it at severity 4 — which is what makes this a readout problem and not a representation-annihilation problem.

Three corollaries, each falsifiable with the released harness — and one already tested. First (*tested, refuted, replaced*): we predicted the floor's class identity would follow the readout's training prior; the imbalanced-prior experiment of §10 refutes this — even with a five-fold-oversampled non-frog prior, the collapse remains frog. The revised claim: the collapse target is a geometry attractor determined by which class prototype sits nearest the collapsed dark manifold, and it should be predictable from embedding distances alone, before any readout is trained. Second, severity-matched augmentation of the readout's training fold should recover most of what the adapted probe recovers, without ever seeing test-time corruptions at training time. Third, any intervention that preserves late-block geometry — whatever its form — should beat any intervention that operates purely in input space; this one is already confirmed at pilot scale, where the adapted readout beats the best enhancement by +15.8 points at s=4 (p=0.007) and +20.0 points at s=5 ($p<10^{-3}$).

For practitioners, the guidance is short. Do not buy float-sensor capture for a frozen DINOv2 pipeline; it purchases nothing. Do not bolt CLAHE onto the front; it will cost you points on clean data. If you adapt, anchor the objective — and expect even that to be second-best to simply recalibrating the readout on data that looks like deployment.

12. Limitations

The honest ones. One backbone family (DINOv2) at two scales; one dataset (CIFAR-10 upsampled from 32px, which flattens the frequency content that the frequency controls probe); a synthetic, globally uniform photometric ladder with no vignetting, auto-exposure dynamics, or real sensor MTF; and a pilot-scale test fold, which makes several paired comparisons under-powered — we flag the ones that are (s=3 adaptation, p=0.14) rather than narrating them as trends. The LoRA ablation runs one seed per configuration; the negative effect is consistent across all six configurations, but per-configuration variance is unmeasured until the full-scale run. Real-darkness validation (ExDark [**?**], SID [**?**]) is the obvious next step and is config-flagged in the harness.

13. Conclusion

Low-light failure in a frozen self-supervised ViT is a readout problem before it is a fine-tuning problem. The damage happens in the continuous image, not at the ADC; it compounds through depth; and the fixed readout answers it by degenerating to a constant class — quietly, in a way per-class accuracy reports as a success story. Retraining the readout on degraded embeddings recovers what enhancement, adapters, and sensor upgrades cannot. Every number in this paper regenerates from one command against committed artifacts, and the statistical machinery behind the claims ships with its own test suite.

A. Reproducibility

Every experiment is a subcommand of `run_experiments.py` (`main_curve`, `cka`, `frequency`, `mechanism`, `quantization`, `lora_ablation`, `mitigation`, `efficiency`, `failure_analysis`, `seed_sensitivity`, `cross_dataset`) driven by `configs/experiments.yaml`; the config SHA-256, package versions, and dataset indices are written to `results/manifest.jsonl` per run. The statistical machinery — unbiased CKA, Wilson and bootstrap intervals, permutation tests, Benjamini–Hochberg correction, the two curve-shape nulls — is covered by 22 unit tests (`tests/run_tests.py`), including a verification that the biased CKA estimator inflates similarity at $n \ll d$ while the estimator used here does not. Noise draws are derived per-image (hash of image bytes and severity), so corruption is bit-reproducible across scripts, processes, and experiment orders. The LoRA ablation, anchored-adapter arm, and STL-10 transfer run on a single GPU via `run_all_colab.py` and fill Tables 7–8.

B. Efficiency Detail

Table 11. Latency (CPU, batch 1, interleaved-round medians over 4 rounds of 25 iterations each).

Arm	Backbone (M)	Trainable (M)	Latency (ms)	Acc (dark, %)
frozen_backbone	22.1	0.00	40.6	56.2
lora_r4	22.4	0.29	49.0	–
lora_r8	22.6	0.59	50.9	–
lora_r16	23.2	1.18	50.7	–